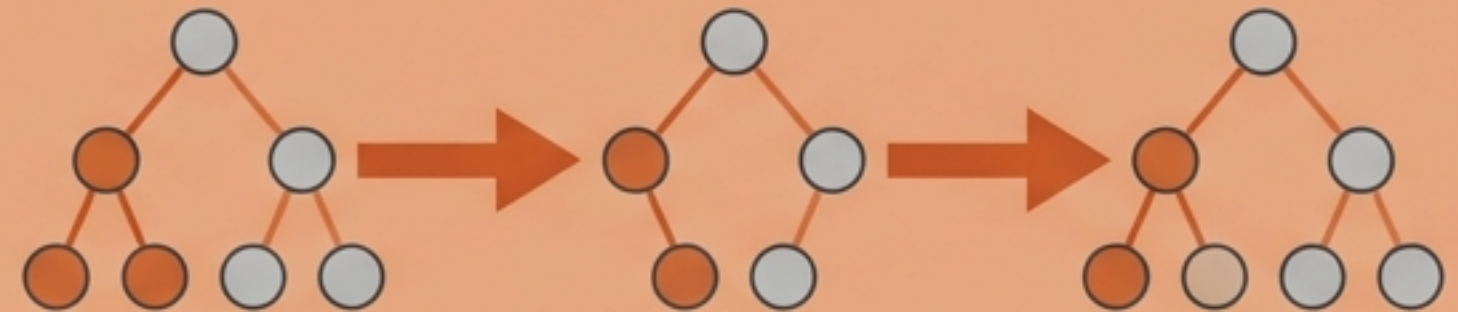
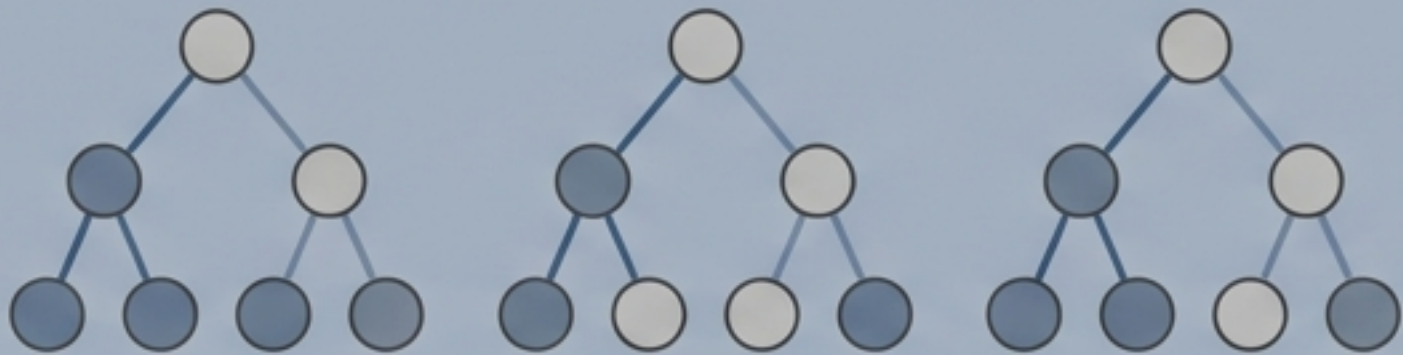
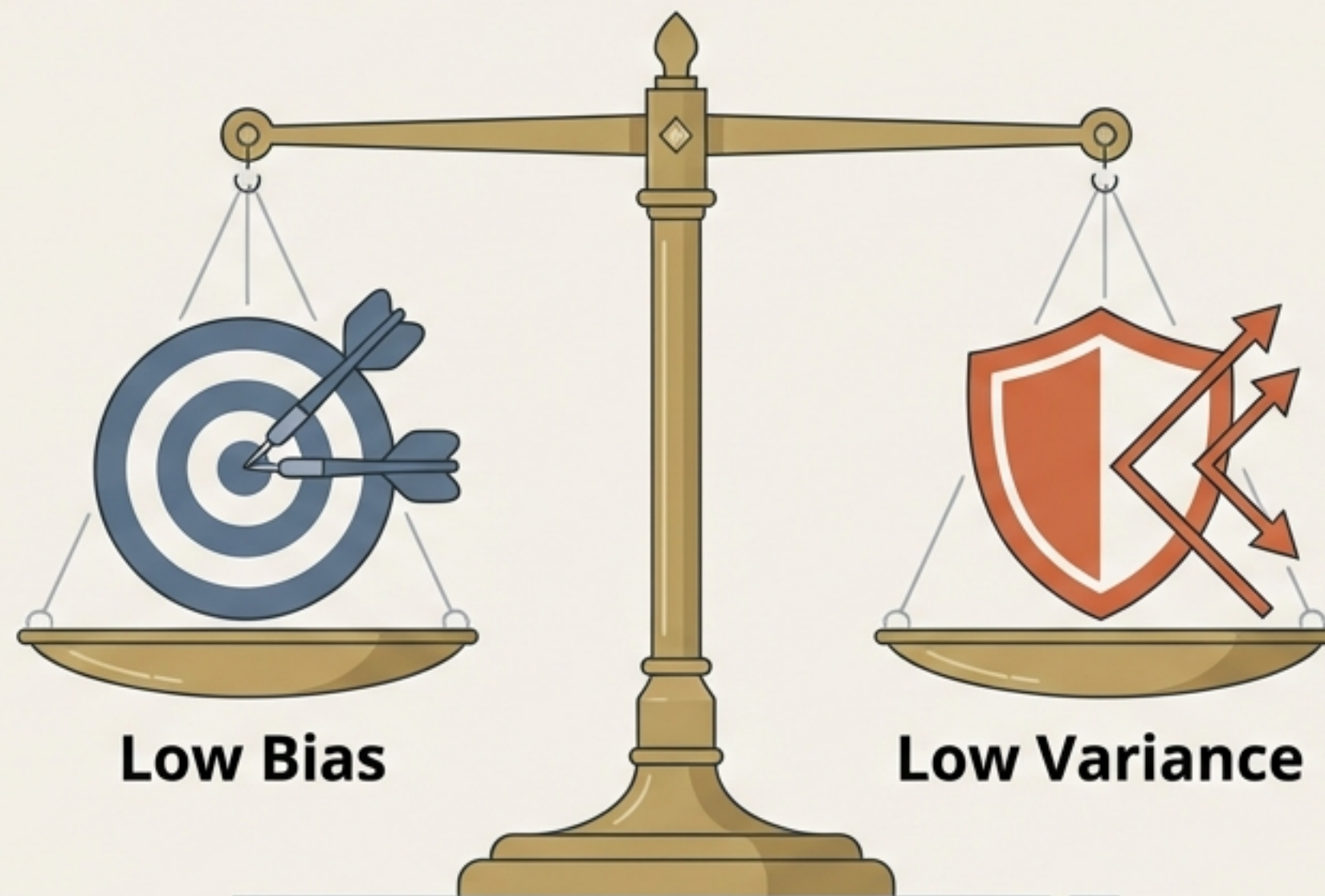


Bagging vs. Boosting



The ultimate goal is achieving low bias and low variance

Excellent, highly accurate results on training data.



Consistent, resilient performance when introduced to new, unseen data changes.

The Reality: Building a single model that perfectly achieves both simultaneously is incredibly difficult.

The inherent conflict of the bias-variance trade-off

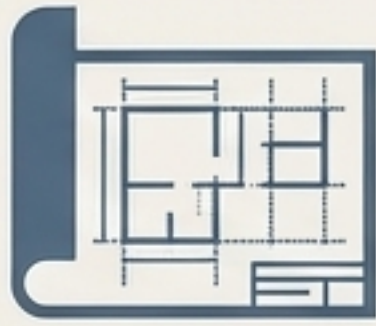


Fixing one problem often exacerbates the other. Decreasing bias naturally increases variance.

To solve this mathematical conflict, the industry relies on three primary techniques: Regularisation, Bagging, and Boosting.

Three fundamental differences separate Bagging and Boosting

While both are ensemble techniques designed to achieve the 'Holy Grail' of low bias and low variance, they operate on completely different philosophies:



1

**Base Model
Characteristics**



2

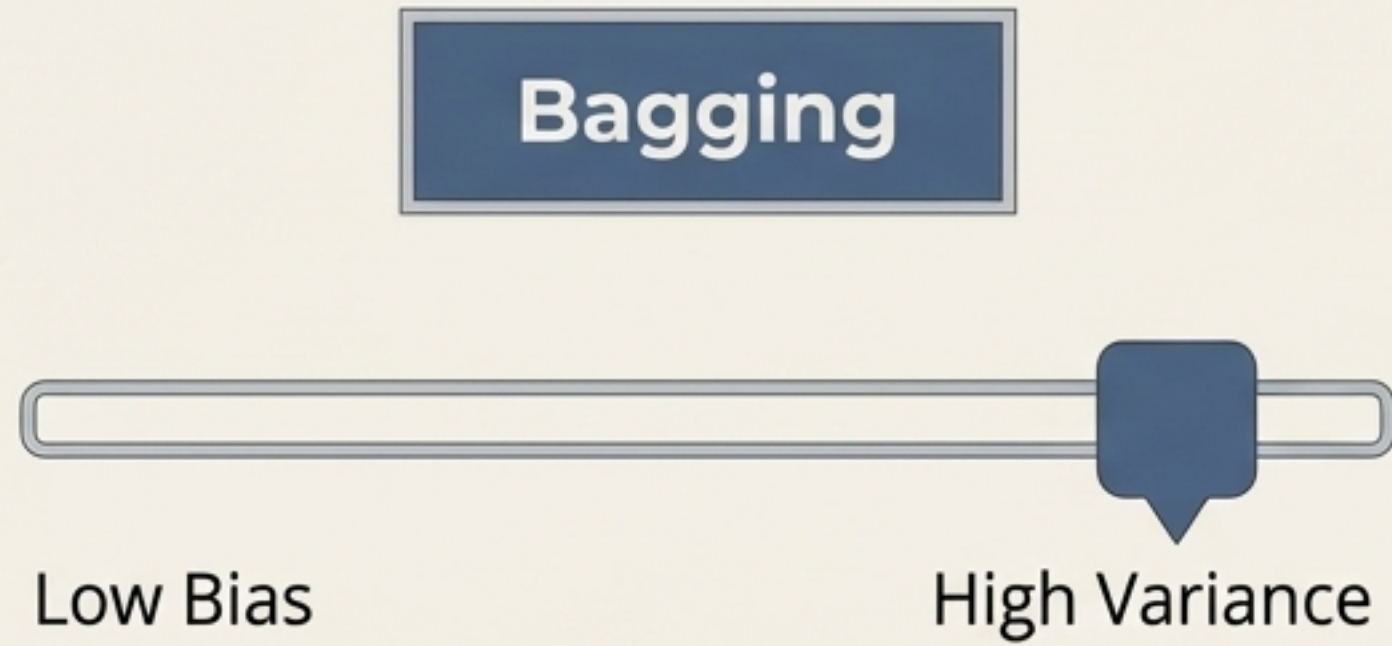
**Training
Methodologies**



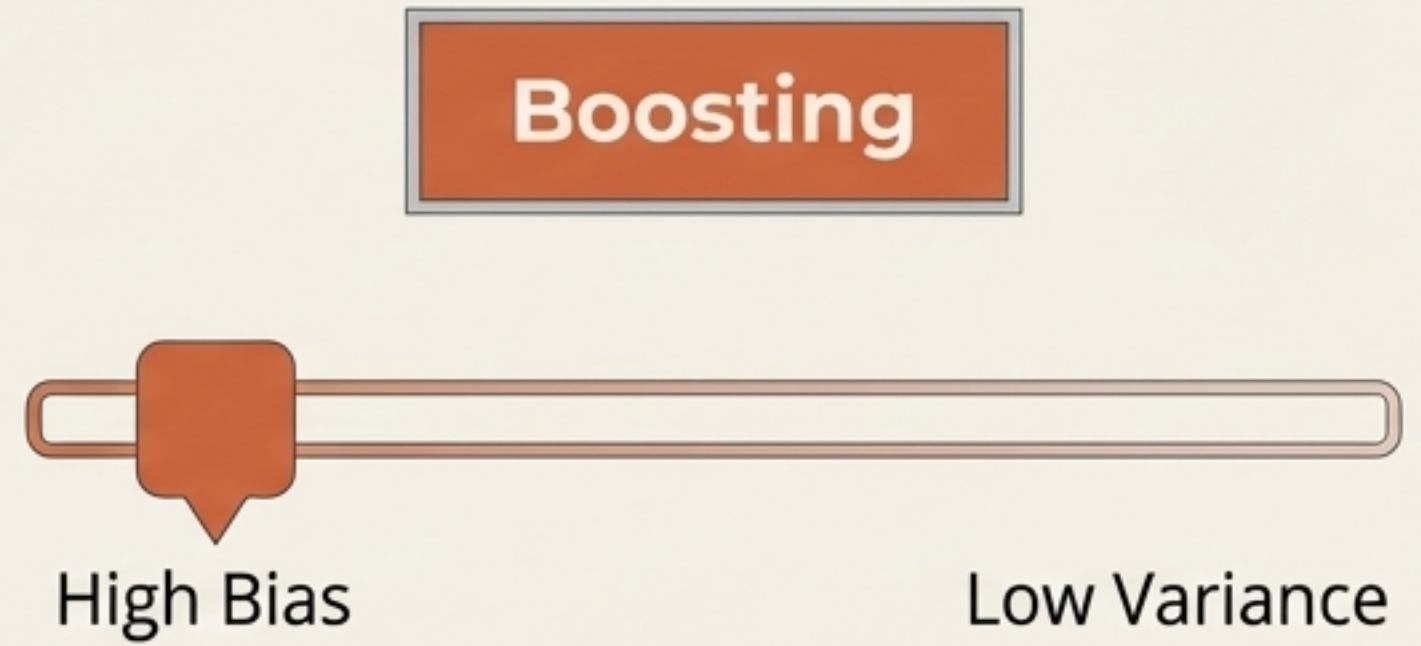
3

**Aggregation
and Weightage**

Bagging and Boosting start from opposite ends of the model spectrum

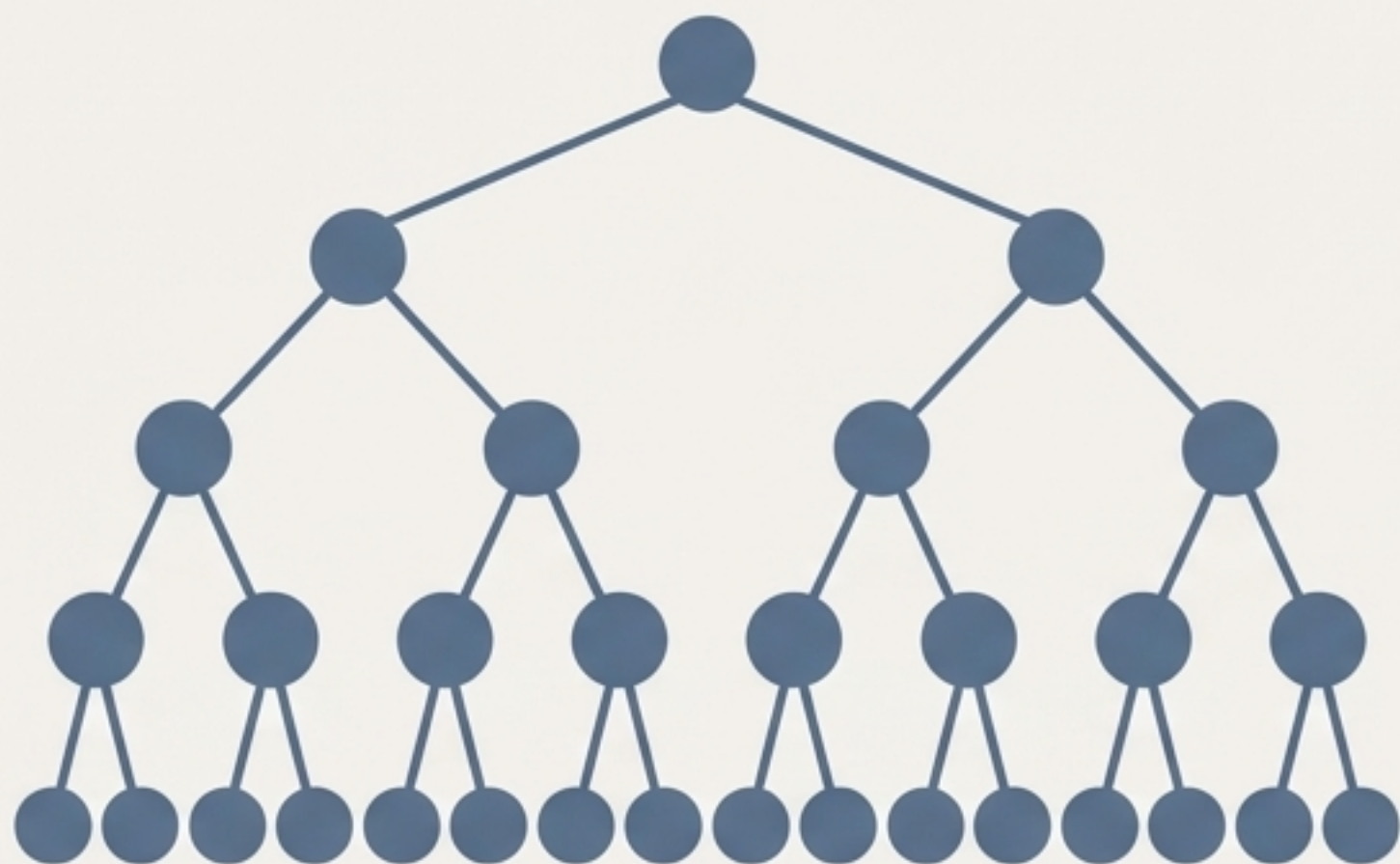


Intentionally selects base models with Low Bias and High Variance. These models are highly accurate on training data but struggle with changes.



Intentionally selects base models with High Bias and Low Variance. These models are overly simplistic and under-perform on training data, but remain unaffected by data changes.

Fully grown trees versus shallow decision stumps



Bagging relies on complex, over-eager models like fully grown decision trees.



Boosting relies on under-prepared, heavily constrained models like shallow decision stumps.

Bagging trains independently while Boosting trains iteratively



Parallel Learning.

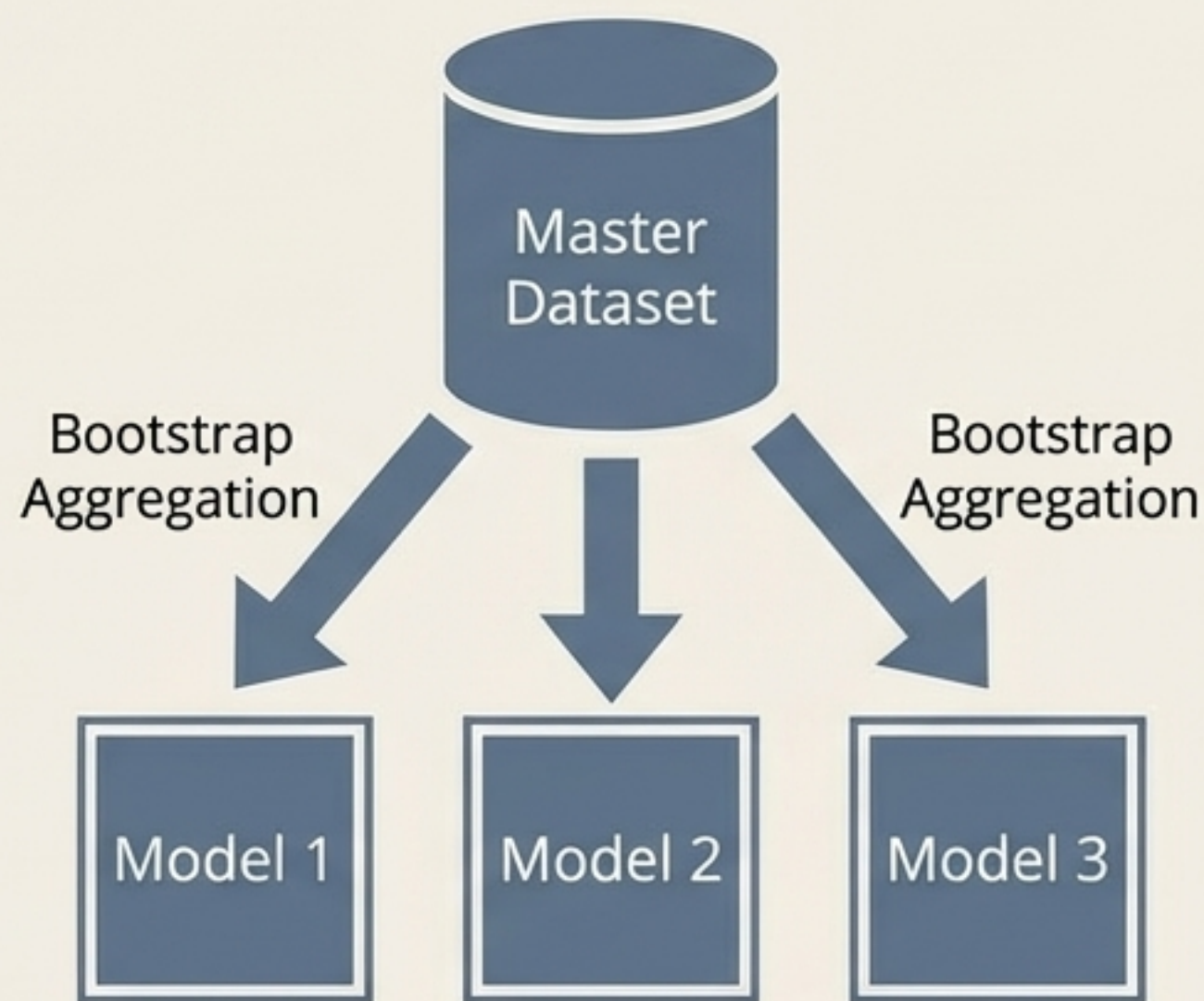
All models are trained simultaneously without communicating with one another



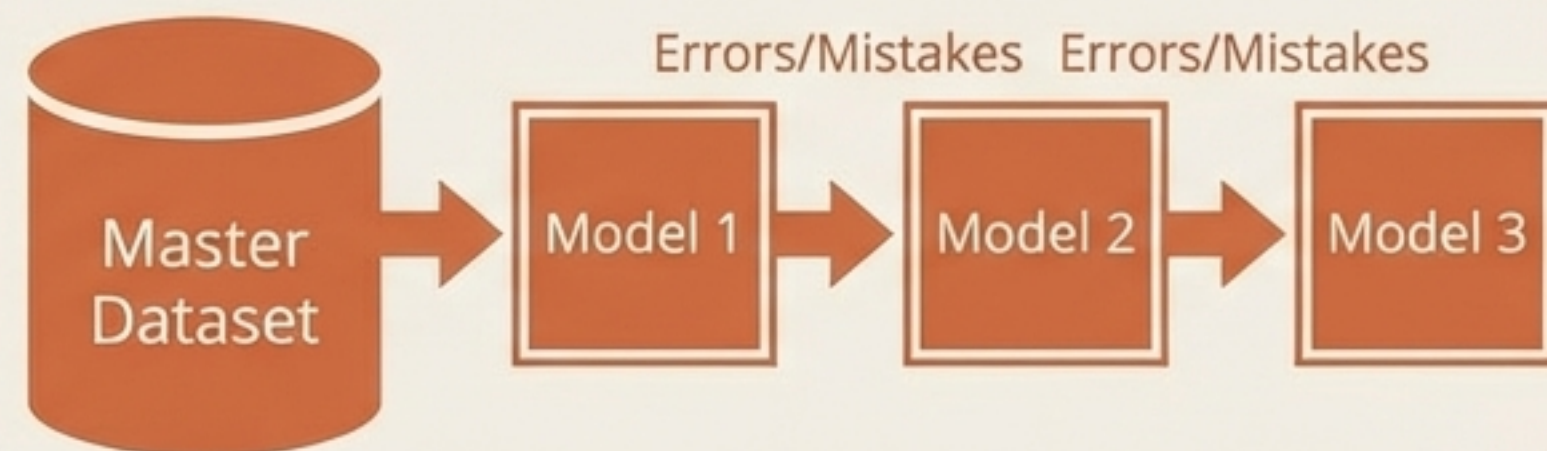
Sequential Learning.

Models are trained one strictly after the other in a chain reaction.

Parallel data sampling versus sequential error passing

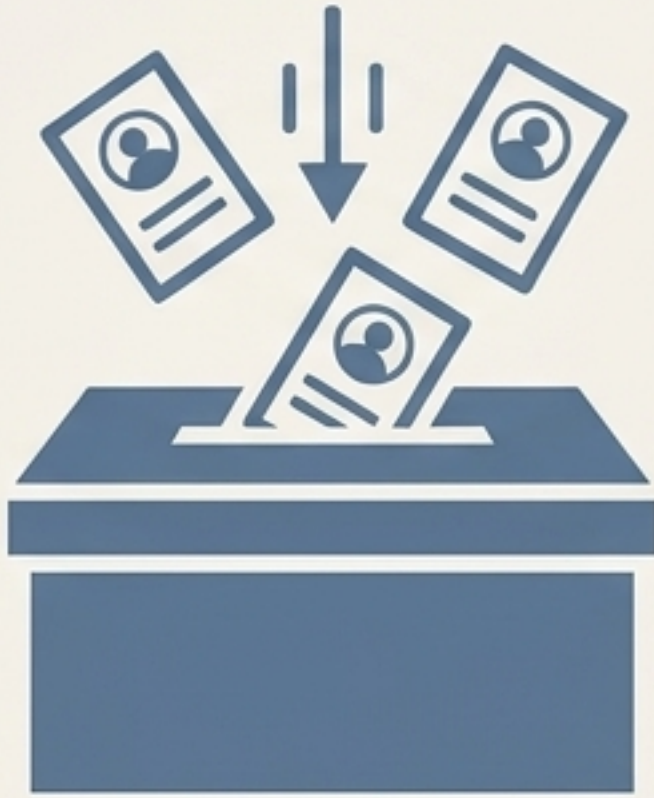


Bagging pulls random portions of data to train separate models at once.



Boosting feeds the whole dataset to the first model, which explicitly teaches the next model what it got wrong.

Final predictions rely on democracy versus meritocracy

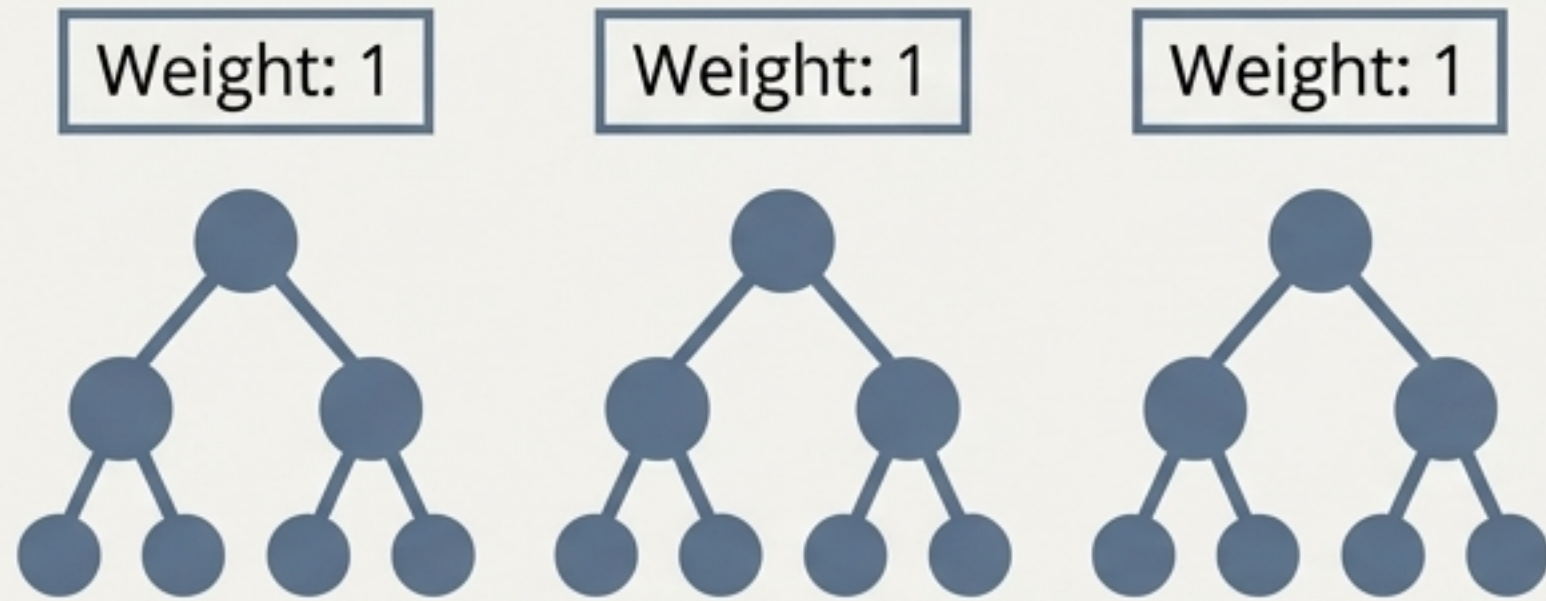


Operates as a pure Democracy.
Every base model is treated as a
total equal in the final decision.



Operates as a Meritocracy.
Individual base models are assigned
varying levels of authority based on
their specific performance.

Equal voting rights versus performance-based influence



In Bagging, every single model gets exactly one equal vote when generating the final prediction.



In Boosting, superior models are mathematically given a louder voice, forcing the algorithm to listen more closely to the better model.

The Ensemble Methods Cheat Sheet

Bagging	Boosting
Base Model: Low Bias / High Variance (e.g., Fully Grown Trees)	Base Model: High Bias / Low Variance (e.g., Shallow Decision Stumps)
Learning Style: Parallel (Independent sampling)	Learning Style: Sequential (Passing errors forward)
Weightage: Equal / Democracy (All 1 vote)	Weightage: Variable / Meritocracy (Weighted by performance)